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1 Abstract

Classical encryption methods, particularly the Caesar cipher, suffer from critical vulnerabilities including trivial key spaces (25 possible keys), susceptibility to frequency analysis, and deterministic outputs that render them inadequate for modern security applications. This coursework addresses these fundamental weaknesses through the development of the Gioco d'azzardo Cipher, a multi-round serial encryption algorithm incorporating 16-round transformations, noise padding, position-based encryption, and Fisher-Yates shuffling. The project encompassed comprehensive research into cryptographic principles, systematic integration of mathematical operations (modular arithmetic, weighted seed derivation supporting 617-digit numeric keys) and logical modifications (bidirectional shifting, variable-length noise padding), followed by rigorous validation through seven distinct test cases covering functional, security, randomness, and robustness testing. The developed algorithm successfully eliminates frequency analysis vulnerabilities, achieves a massive key space of approximately 10^{617} possibilities, implements non-deterministic cipher text generation, and demonstrates capability to process extended character sets while maintaining data integrity. This enhanced cipher provides practical security for educational environments, low-stakes personal applications, gaming software, and CTF competitions, while clearly establishing boundaries for inappropriate high-security applications including financial systems, healthcare data, and regulatory-compliant scenarios.

2 Introduction

Cryptography, the science of securing information through mathematical transformations, has evolved from ancient techniques to sophisticated modern algorithms addressing contemporary cyber threats. The Caesar cipher, despite its historical significance as one of the earliest encryption methods, possesses fundamental vulnerabilities rendering it obsolete for practical security applications. This coursework undertakes the systematic

enhancement of the Caesar cipher through integration of contemporary cryptographic principles including multi-round serial transformation, seed-based key derivation, dynamic character substitution, and probabilistic encryption mechanisms. The resulting Gioco d'azzardo Cipher addresses core weaknesses while maintaining computational efficiency suitable for educational purposes and low-stakes applications, demonstrating the evolution from classical to modern cryptographic design principles.

2.1 Aim

To research, design, develop, test, and evaluate an enhanced Caesar cipher based cryptographic algorithm incorporating multi-round transformations and position dependent encryption to address classical vulnerabilities.

2.2 Objectives

- To conduct comprehensive research on cryptographic systems including symmetric and asymmetric encryption, analyzing the Caesar cipher's mechanisms and critical vulnerabilities.
- To design and implement mathematical modifications including modular arithmetic, position-based seed derivation, and weighted seed calculation supporting 617-digit numeric keys.
- To design and implement logical modifications including 16-round serial transformation, Fisher-Yates shuffle algorithm, bidirectional shifting, and variable-length noise padding.
- To develop complete encryption and decryption algorithms with numeric-only seed validation enabling Diffie-Hellman key exchange compatibility.
- To create comprehensive documentation including step-by-step encryption walkthroughs and visual flowchart representation of the algorithmic process.

- To execute rigorous testing through multiple test cases covering functional validation, security testing, randomness verification, and robustness assessment.
- To critically evaluate the algorithm by analyzing strengths (frequency analysis elimination, massive key space, avalanche effect) and weaknesses (weak PRNG, cipher text expansion, lack of authentication).

3 Cryptographic Systems

3.1 Computer Security

Cybersecurity, often known as computer security, is a system that guards against theft, abuse, unauthorized use, and damage to important data and information contained in a computer system. It is an operating system that is configured for computers in order to safeguard the information and data held within the specific system. It refers to the security system established in computer systems to prevent unauthorized access, abuse, and theft of critical data and information kept on the computer. Applications and systems are shielded from other harmful actions by computer security in cyber security (Sharma & www.knowledgehut.com, 2025).

Since the computer hardware is already secured in a locker or other secure location, they are often employed to prohibit computer software. Therefore, these computer security systems provide protection for the computer program, which is susceptible. These intricate security procedures and strategies make it difficult for hackers to obtain vital data. These systems are designed to protect sensitive and important data and information that, if compromised, might result in significant losses for an individual or an organization (Sharma & www.knowledgehut.com, 2025).

3.2 Types of Computer Security

3.2.1 Application Security

Application security is the term used when security measures are added within the first phase of the development process. It may effectively defend your computer system from cyber security risks including illegal access and data leaks. Additionally, it can assist your computer system in defending against denial of service assaults and SQL intrusions (Sharma & www.knowledgehut.com, 2025).

3.2.2 Information Security

A subset of cyber security known as information security focuses on the methods and strategies developed to guarantee computer security. Information security was created as a process to safeguard computer systems' availability, confidentiality, and integrity against data theft, unauthorized access, damage, and destruction (Sharma & www.knowledgehut.com, 2025).

3.2.3 Endpoint Security

Hackers and cybercriminals may readily take advantage of any human error. In any company, end users face a significant security risk. Due to their ignorance of IT security and regulations, end users become prey to cybercrimes. They may unintentionally provide cybercriminals access to their computer systems because they are unaware (Sharma & www.knowledgehut.com, 2025).

3.2.4 Internet Security

The most modern form of computer security that has experienced a recent surge is internet security. It is a technique for developing the ideal set of guidelines and procedures

to stop any unauthorized usage or damage to computer systems that are directly connected to the internet (Sharma & www.knowledgehut.com, 2025).

3.3 CIA Triad

The CIA security triangle concept is built around the ideas that information availability, confidentiality, and integrity are critical to the operation of the company. The cyber security paradigm known as the CIA triangle divides these three ideas into distinct focus points. This distinction is beneficial because it allows security teams to identify many approaches to each issue. When all three criteria are met, the organization's security structure should be more forceful and better equipped to deal with threat situations (Prakash & www.knowledgehut.com, 2022).

Components of the CIA Triad

3.3.1 Confidentiality

Confidentiality refers to an organization's attempts to ensure that information is kept private or hidden. In order to do this, information access must be restricted in order to stop data from being shared without authorization, whether on purpose or accidentally. Ensuring that anyone without the necessary authorization cannot access assets that are crucial to your organization is a crucial part of protecting confidentiality. On the other hand, an efficient system also guarantees that the appropriate rights are granted to those who require access (www.fortinet.com, 2026).

3.3.2 Integrity

Ensuring that your data is reliable and unaltered is part of maintaining integrity. Your data's integrity is only preserved if it is genuine, accurate, and trustworthy. For instance, information about top management that your organization posts on its website must be

accurate. Inaccurate information might give the impression that your company is unreliable to people who are looking for information. Someone who has a stake in harming your company's image could attempt to hack your website and change the executives' titles, photos, or descriptions in order to harm the company's reputation overall or their own (www.fortinet.com, 2026).

3.3.3 Availability

Data is frequently meaningless unless it is accessible to both the organization's employees and the clients they serve, even if it is kept private and intact. This implies that applications, networks, and systems must be operating when and as they should. Additionally, those who have access to certain information must be able to use it when necessary, and accessing the data shouldn't take an excessive amount of time (www.fortinet.com, 2026).

Availability will be jeopardized, for instance, if there is a power outage and no disaster recovery plan in place to assist users in regaining access to vital services. Additionally, users may be unable to access their workstations and other devices that deliver business-critical information or apps if a natural disaster, such as a flood or even a strong blizzard, prevents them from coming to the workplace. Deliberate acts of sabotage, such as ransomware or denial-of-service (DoS) assaults, can potentially jeopardize availability (www.fortinet.com, 2026).

3.4 Cryptography

The science of encrypting and decrypting data using mathematics is known as cryptography. Sensitive information can be stored or sent over unsecure networks (such as the Internet) using cryptography so that only the intended receiver can access it. Cryptanalysis is the study of analyzing and cracking encrypted communication, whereas cryptography is the science of data security. Analytical thinking, the use of mathematical tools, pattern recognition, patience, perseverance, and good fortune are all intriguing

components of classical cryptanalysis. Another name for cryptanalysts is attackers. Cryptography and cryptanalysis are both included in cryptology (Corporation, 2002).

Plaintext, also known as clear text, is information that can be read and comprehended without any further steps. Encryption is the process of hiding plaintext's content by disguising it. Cipher text is the unintelligible jargon produced by encrypting plaintext. Even those who are able to view the encrypted material can be prevented from accessing it by using encryption. Decryption is the process of changing cipher text back to plaintext (Corporation, 2002).

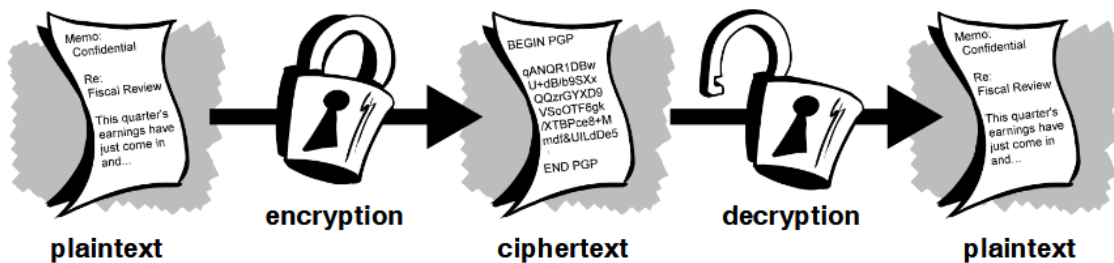


Figure 1:Encryption and Decryption

A mathematical function employed in the encryption and decoding process is known as a cryptographic algorithm, or cipher. To encrypt the plaintext, a cryptographic technique is used in conjunction with a key, which might be a word, number, or phrase. With multiple keys, the same plaintext can be encrypted into a distinct cipher text. The strength of the cryptographic method and the key's secrecy are the only factors that determine how secure encrypted data is. A cryptosystem is made up of a cryptographic algorithm, all potential keys, and all the protocols that enable it (Corporation, 2002).

3.4.1 Symmetric Cryptography

A secret key in symmetric key cryptography, often called private key cryptography, can be shared between the sender and the recipient of a communication or kept by one

individual. Both the sender and the recipient must have a copy of the secret key if private key cryptography is used to transmit confidential messages between two parties (Naser & www.eajournals.org, 2021).

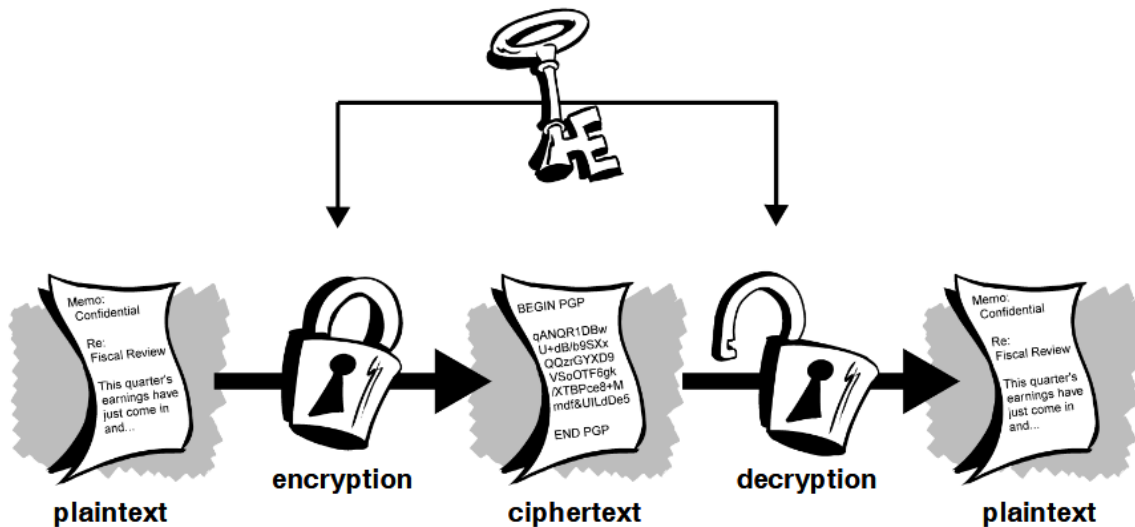


Figure 2: Symmetric Encryption

Conventional encryption has advantages. It moves quickly. When encrypting data that is not going anywhere, it is extremely helpful. However, because safe key distribution may be challenging, using traditional encryption alone to deliver protected data can be quite costly. Conventional encryption requires that a sender and recipient agree on a key and maintain it confidential in order to communicate safely. They must rely on a courier, the Bat Phone, or another secure communication tool if they are physically separated in order to avoid the secret key being revealed during transmission. All data encrypted or authenticated with that key can thereafter be read, altered, and forged by anyone who overhears or intercepts the key while it is in transit (Corporation, 2002).

3.4.2 Asymmetric Cryptography

One key encrypts the data in the two key system (also called the public key system), while another mathematically related key decrypts it. An encrypted communication is sent by a computer using a selected private key that is only known by the sender and is never shared. The sender and recipient can authenticate each other and safeguard the message's confidentiality by using this public-key cryptography technique (Naser & www.eajournals.org, 2021).

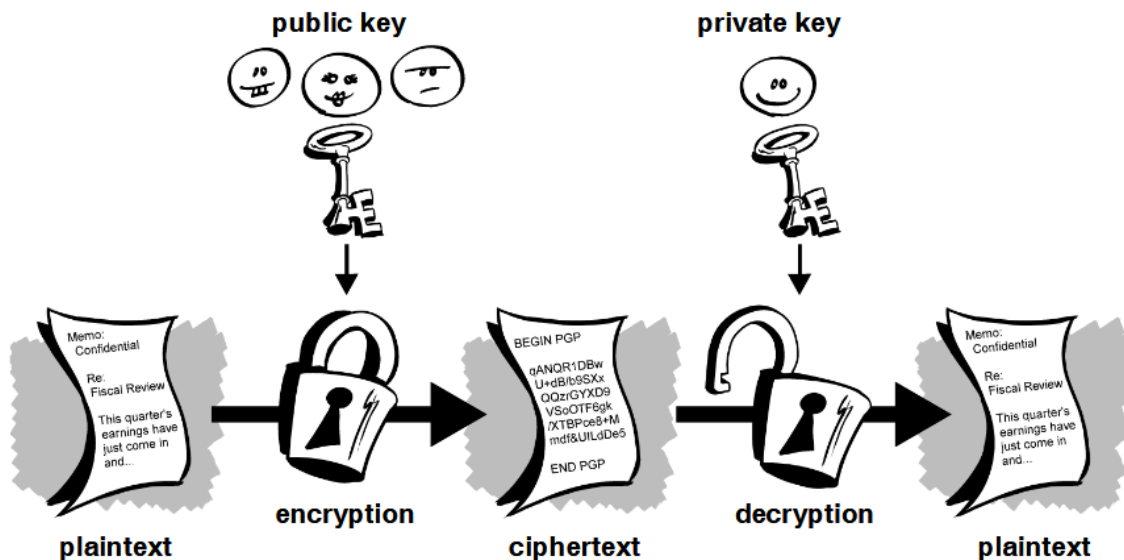


Figure 3: Asymmetric Cryptography

The private key cannot be calculated from the public key. Information can be encrypted but cannot be decrypted by anybody with a public key. The information can only be decrypted by the individual who possesses the matching private key. The main advantage of public key cryptography is that it enables safe message transmission between individuals without any prior security setup. There is no longer a requirement for sender and recipient to exchange secret keys via a secure channel; only public keys are used in all communications, and private keys are never shared or transferred (Corporation, 2002).

3.4.3 Key Terminologies

Plaintext

Plaintext in cryptography often refers to unencrypted data or the original, comprehensible message that someone wants to send to another person, supplied into the algorithm as input (Naser & www.eajournals.org, 2021).

Key

A key in cryptography is a piece of data (a parameter) that establishes how a cryptographic algorithm operates. Depending on the decryption technique, a key for encryption algorithms defines how plaintext is converted into cipher text and vice versa. Other cryptographic methods, such message authentication codes and digital signature techniques, also use keys to define transformations (Naser & www.eajournals.org, 2021).

Secret Key

The encryption algorithm also receives the secret key. The algorithm and the plaintext have no bearing on the value of the key. Depending on the particular key being used at the moment, the algorithm will yield a different result. The key determines the precise substitutions and changes that the program makes (Naser & www.eajournals.org, 2021).

Cipher Text

Cipher text, also known as cypher text, is information that has been encrypted or encoded because it contains a version of the original plaintext that cannot be decrypted by a machine or person without the right cipher. This is the output, which is a jumbled message. Both the secret key and the plaintext are necessary (Naser & www.eajournals.org, 2021).

Encryption

Encryption is the process of changing plain text into cipher text. To convey private communications via an unsecure connection, cryptographers employ a variety of encryption techniques. An encryption algorithm and a key are necessary for the encryption process. The technique used to encrypt the data is known as an encryption algorithm. On the sender's end, encryption takes place (Naser & www.eajournals.org, 2021).

Decryption

Decryption is the name for the encryption process in reverse. It is the process of turning plain text from cipher text. In order to recover the original message from an unintelligible communication, or cipher text, cryptographers employ decryption techniques on the receiving end. A decryption algorithm and a key are necessary for the decryption process. An algorithm for decryption refers to the technique employed in decryption. The encryption and decryption algorithms are often the same, except they are reversed (Naser & www.eajournals.org, 2021).

3.5 History

The usage of encryption dates back to the Babylonian Era, around 3000 B.C. Because of the recent widespread use of the Internet and the dramatic increase in the amount of information people come into contact with in their daily lives, the settings in which encryption technologies are applied and implemented have increased, and they are now used all around us in our daily lives. However, encryption technologies evolved as they were used in military and political settings (Thawte & adgrafics.net, 2013).

The earliest known example of encryption is said to be hieroglyphics, or pictograms employed in ancient Egypt, which were written on a stele around 3000 B.C. Before the

Rosetta Stone was discovered and studied in the 19th century, it was thought that it would never be feasible to understand hieroglyphics (Thawte & adgrafics.net, 2013).

In the sixth century B.C., the city-state of Sparta in ancient Greece employed a type of encryption known as the "scytale cipher." The inscription was inscribed on the parchment strip along the cylinder's long axis after a cylinder of a specific diameter was wrapped around it. The encryption technique was created so that the receiver could read it by encircling a parchment strip around a cylinder of the same diameter (Thawte & adgrafics.net, 2013).

4 Caesar Cipher

4.1 Background

The Caesar cipher, which first appeared in the first century B.C., got its name from Julius Caesar's frequent use of it. It is one of the most well-known encryption techniques among the many that have developed during the lengthy history of encryption (Thawte & adgrafics.net, 2013).

The Caesar cipher technique of encryption substitutes a letter that is situated a predetermined number of places farther down the language's alphabet for each letter in the original text. Each letter in the text will be replaced by a letter that is, for example, three letters lower in the sender's and recipient's alphabets (Thawte & adgrafics.net, 2013).

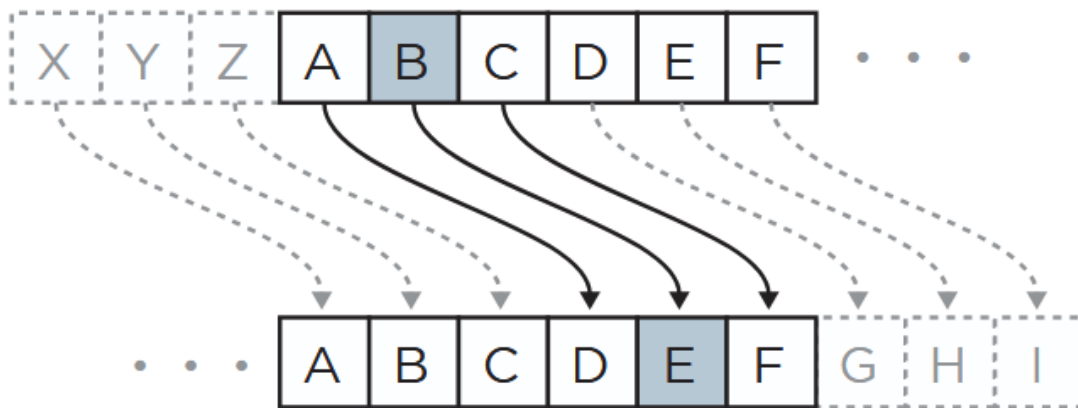


Figure 4: Ceaser Cipher

The Caesar cipher is frequently referred to as a "shift cipher" since it required character shifting. Texts encrypted using the Caesar cipher may be decoded by attempting 26 patterns if the alphabet has 26 letters. However, the sequence can be randomly rearranged instead of just shifting the characters by a predetermined number of places in the alphabet. This greatly increases the number of possible patterns (in the case of a 26-letter alphabet: $26 \times 25 \times 24 \times \dots = 400,000,000,000,000,000,000,000,000$ patterns!) and makes decryption much more challenging (Thawte & adgraphics.net, 2013).

Two alphabets can be aligned to indicate the transformation; the cipher alphabet is the plain alphabet rotated by a certain number of places to the left or right. For example, the following Caesar cipher uses a three-place left rotation (the shift parameter, in this case, 3, is utilized as the key (Babylon & www.uobabylon.edu.iq, 2026).

Plain: ABCDEFGHIJKLMNOPQRSTUVWXYZ

Cipher: DEFGHIJKLMNOPQRSTUVWXYZABC

In order to encrypt a message, one must look up each letter in the "plain" line and write the matching letter in the "cipher" line. The process of deciphering is reversed (Babylon & www.uobabylon.edu.iq, 2026).

Cipher text: WKH TXLFN EURZQ IRA MXPSV RYHU WKH ODCB GRJ

Plaintext: the quick brown fox jumps over the lazy dog

$$E_n(x) = (x + n) \bmod 26$$

(Encryption Phase with shift n)

$$D_n(x) = (x - n) \bmod 26$$

(Decryption Phase with shift n)

Figure 5: Ceaser Cipher Formula

Modular arithmetic may also be used to express the encryption by first converting the letters into integers in accordance with the scheme, A = 0, B = 1,..., Z = 25. $E_n(x) = (x+n) \bmod 26$ is the mathematical expression for encryption of a letter x by a shift n, while $D_n(x) = (x-n) \bmod 26$ is the mathematical expression for decryption. The modulo operation is defined in several ways. The outcome falls inside the range of 0...25; that is, we must subtract or add 26 if x+n or x-n do not fall within this range. In contrast to polyalphabetic substitution, the cipher is categorized as a kind of monoalphabetic substitution since the replacement stays the same throughout the message (Babylon & www.uobabylon.edu.iq, 2026).

4.2 Advantages

- Beginners may learn about encryption since it is simple to use and apply (www.geeksforgeeks.org, 2025).
- Can be physically accomplished, for example, using a scytale a series of cards or spinning disks which might be helpful in some circumstances (www.geeksforgeeks.org, 2025).

- Just needs a minimal amount of previously shared data (www.geeksforgeeks.org, 2025).

4.3 Disadvantages

- It is vulnerable to contemporary decryption techniques (www.geeksforgeeks.org, 2025).
- Susceptible to known-plaintext attacks, in which the same communications may be accessed by an attacker in both encrypted and unencrypted forms (www.geeksforgeeks.org, 2025).
- It is susceptible to a brute force attack because of the limited number of available keys, which makes it simple for an attacker to attempt every key until the right one is discovered (www.geeksforgeeks.org, 2025).
- Fails to offer confidentiality, integrity, and authenticity within a message (www.geeksforgeeks.org, 2025).

5 Development of New Cipher

5.1 Gioco d'azzardo Cipher

A Multi-Round Substitution Cipher with Cryptographic Security

5.2 Necessity of Modifications

Limitations of Classical Caesar Cipher

The traditional Caesar cipher suffers from critical security vulnerabilities that make it unsuitable for modern cryptographic applications:

1. Trivial Key Space

- Only 25 possible keys (shifts of 1-25)
- An attacker can brute-force all possibilities in seconds
- Example: "HELLO" with shift 3 → "KHOOR" (easily breakable)

2. Frequency Analysis Vulnerability

- Letter frequencies remain unchanged (e.g., 'E' is still the most common)
- Cryptanalysts can identify patterns by analysing letter distribution
- Single substitution makes it transparent to statistical attacks

3. Deterministic Output

- Same plaintext always produces identical cipher text with the same key
- Enables known-plaintext and chosen-plaintext attacks
- No semantic security

4. Limited Character Support

- Traditional implementation only handles A-Z (26 letters)
- Cannot encrypt numbers, punctuation, or special characters
- Inadequate for modern digital communication

5.3 Mathematical and Logical Modifications

Mathematical Modifications

1. Modular Arithmetic

- $(\text{index} \pm \text{shift}) \% \text{charset_length}$
- **Purpose:** Ensures indices wrap around the charset, creating a circular substitution space

- **Security benefit:** Eliminates boundary patterns and creates uniform distribution

2. Position-Based Seed Derivation

- `Character_PRNG = Random(base_seed + position)`
- **Purpose:** Each character position uses a different PRNG state
- **Security benefit:** Prevents identical characters from encrypting the same way

3. Weighted Seed Calculation

- $\text{base_seed} = \sum(\text{ord}(c) \times (i+1))$ for each character c at position i
- **Purpose:** Creates unique numeric seed from any phrase
- **Security benefit:** Makes seed space astronomically large (supports 617-digit seeds)

4. Random Range Generation

- Dice: `randint(1, 6)` provides shift magnitude
- Noise: `randint(2, 5)` provides variable padding
- **Purpose:** Introduces controlled randomness
- **Security benefit:** Makes patterns unpredictable while maintaining decryption capability

Logical Modifications

Multi-Round Iteration

- 16 sequential transformation rounds
- **Purpose:** Achieves confusion and diffusion (Shannon's principles)
- **Security benefit:** Creates avalanche effect—small input changes cascade through rounds

Bidirectional Shifting

- Random choice between 'R' (right) and 'L' (left) shift
- **Purpose:** Doubles the transformation space

- **Security benefit:** Prevents pattern recognition from unidirectional shifts

Noise Padding with Variable Length

- Random 2-5 noise characters before each real character
- **Purpose:** Obscures message structure and length
- **Security benefit:** Same plaintext produces different cipher texts each time; prevents frequency analysis

Deterministic Randomness (PRNG Seeding)

- Uses seeded PRNGs instead of true randomness
- **Purpose:** Makes encryption reproducible for decryption
- **Security benefit:** Provides cryptographic randomness while maintaining symmetric key property

5.4 New Methodology Implied

The enhanced cipher introduces several sophisticated mechanisms:

5.4.1 1. Seed-Based Key Derivation

- Uses a shared seed phrase (supporting up to 617 digits) instead of a simple shift number
- Generates a master PRNG (Pseudo-Random Number Generator) from the seed
- Creates position-specific PRNGs for each character, ensuring different encryption for each position

5.4.2 2. Multi-Round Serial Transformation

- Applies 16 sequential rounds of encryption (industry standard for high security)

- Each round uses a different shuffled charset and random parameters
- Creates an "avalanche effect" where small input changes drastically alter output

5.4.3 3. Noise Padding

- Adds 2-5 random noise characters before each real character
- Adds random noise at the end of the message
- Makes cipher text length variable and unpredictable
- Prevents pattern analysis and makes identical plaintexts produce different cipher texts

5.4.4 4. Dynamic Character Substitution

- Uses a 91-character charset (letters, digits, punctuation, spaces)
- Shuffles the charset differently for each round
- Applies random directional shifts (left or right)
- Uses random shift amounts (1-6 positions)

5.5 Encryption

ONE ROUND of encryption with plaintext **"HI"**:

Plaintext: "HI"

Seed Phrase: "secret123"

Charset: " abcdefghijklmnopqrstuvwxyz...XYZ0123456789!@#\$..." (91 characters)

Step-by-Step Process for Character 'H' (Position 0):

Step 1: Noise Padding

Master PRNG generates: num_noise = 3

Add 3 random noise characters: "x\$9"

Current output: "x\$9"

Step 2: Initialize Character-Specific PRNG

Base seed = sum of (character_code × position) from "secret123"

Character PRNG = Random(base_seed + 0) // position 0 for 'H'

Step 3: Round 1 Transformation

(3a) Shuffle the Charset

Character PRNG shuffles the 91-character charset

Original: " abcdefgh...GHIJKLMN..."

Shuffled: "m3!aX9...pHq7K2L..." (example shuffled order)

3b) Generate Round Parameters

Dice roll = 4 (random number 1-6)

Coin flip = 'R' (random choice of 'R' or 'L')

(3c) Find 'H' in Shuffled Charset

In shuffled charset "m3!aX9...pHq7K2L..."

'H' is at index position 42 (example)

(3d) Apply Transformation

Since coin = 'R' (Right shift):

New index = $(42 + 4) \% 91 = 46$

Character at index 46 in shuffled charset = 'q'

Current value changes: 'H' $\rightarrow$ 'q'

After 1 Round:

'H' has been transformed to 'q'

Full output so far: "x\$9q"

Step-by-Step Process for Character 'l' (Position 1):

Step 1: Noise Padding

Master PRNG generates: num_noise = 4

Add 4 random noise characters: "@k2%"

Current output: "x\$9q@k2%"

Step 2: Round 1 Transformation for 'l'

Character PRNG = Random(base_seed + 1) // position 1

Shuffled charset (different from 'H'): "7b&Yz...lqr..."

Dice = 2

Coin = 'L' (Left shift)

'l' at index 38 in this shuffled charset

New index = $(38 - 2) \% 91 = 36$

Character at index 36 = 'r'

'l' → 'r'

After encrypting both characters (1 round only):

Output: "x\$9q@k2%r" + (final noise)

Final noise: "!7" (2 characters)

Complete cipher text (1 round): "x\$9q@k2%r!7"

5.6 Decryption

Decryption reverses the process using the same seed

Decrypting "x\$9q@k2%r!7" to recover 'H':

Step 1: Skip Noise Characters

Master PRNG (same seed) generates: num_noise = 3

Skip first 3 characters: "x\$9"

Current position: index 3, character 'q'

Step 2: Initialize Character-Specific PRNG

Character PRNG = Random(base_seed + 0) // character count 0

Step 3: Generate Round Parameters (Same as Encryption)

Shuffled charset: "m3!aX9...pHq7K2L..." (identical to encryption)

Dice = 4

Coin = 'R'

Step 4: Reverse the Transformation

Find 'q' in shuffled charset at index 46

Since coin was 'R' in encryption, we reverse with LEFT:

New index = $(46 - 4) \% 91 = 42$

Character at index 42 = 'H'

Recovered: 'q' → 'H' ✓

Decrypting to recover 'l':

Step 1: Skip Noise

Master PRNG generates: num_noise = 4

Skip 4 characters: "@k2%"

Current position: character 'r'

Step 2: Reverse Transformation

Shuffled charset: "7b&Yz...lqr..."

Find 'r' at index 36

Coin was 'L', so reverse with RIGHT:

New index = $(36 + 2) \% 91 = 38$

Character at index 38 = 'l'

Recovered: 'r' $\rightarrow$ 'l' ✓

Final Decrypted Output: "HI"

5.7 Flowchart

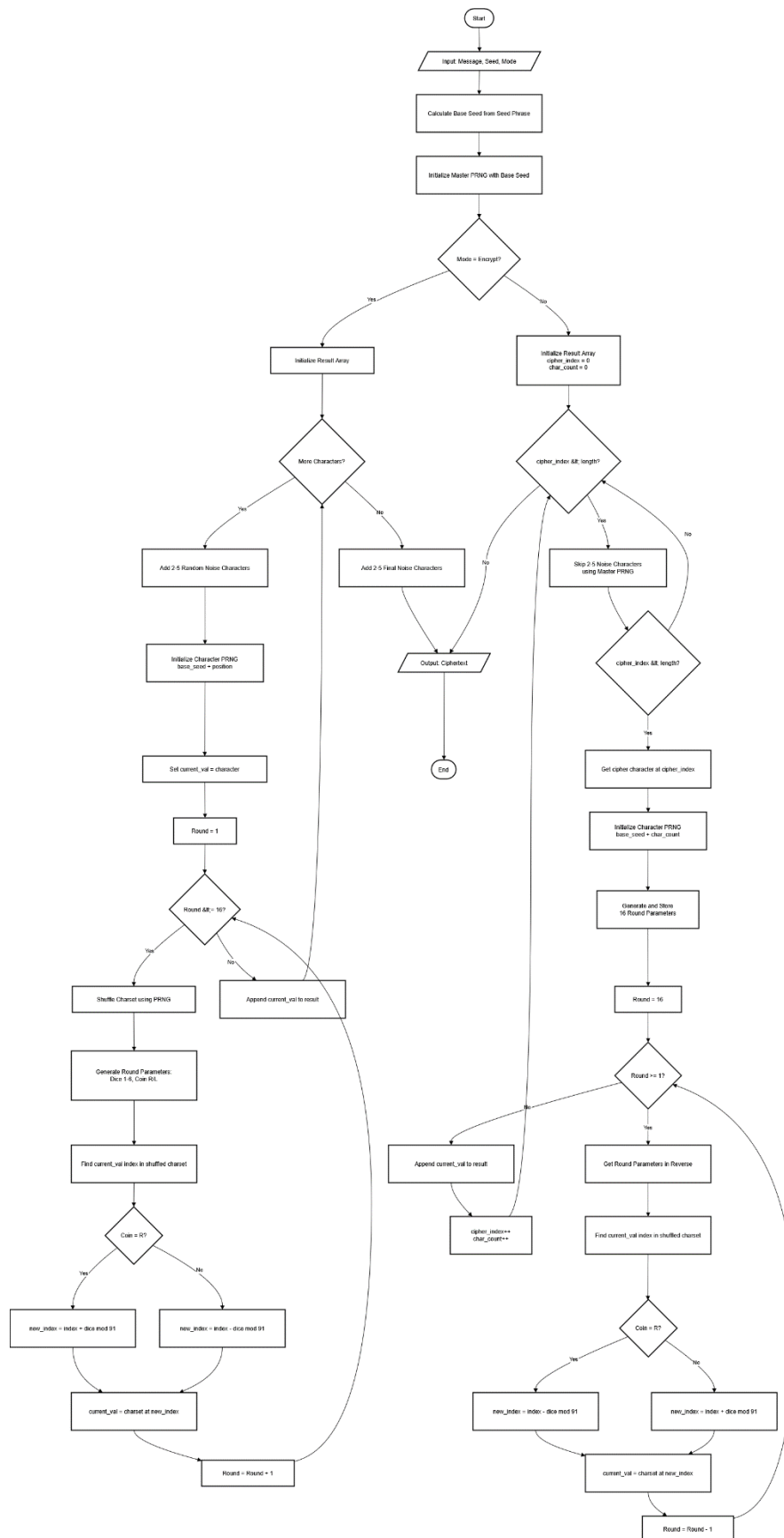


Figure 6:Flowchart

6 Testing

6.1 Test 1

Test Name	Basic Encryption-Decryption Test
Objective	Verify that a simple message can be encrypted and then successfully decrypted back to the original plaintext using a numeric seed
Action	1. Enter plaintext "HELLO" 2. Enter numeric seed "555" 3. Encrypt the message 4. Decrypt the resulting cipher text using the same seed "555"
Expected Result	Decrypted text matches original plaintext "HELLO" exactly
Actual Result	Decrypted text: "HELLO"
Test Status	Successful

Table 1:Test 1

Images below provide evidence:

Images below provide evidence:

```
C:\Users\Nitro v\AppData\Loc x + v
  ◆ GIOCO D'AZZARDO ◆
Multi-Round Serial Cipher (16 Rounds)

[1] Encrypt [2] Decrypt [3] Exit
Select Option: 1
Enter text: SECRET
Enter Numeric Seed (617-digit number): 98765

--- ENCRYPTED RESULT ---
NRu>0eCw$k/},oMzWUI@^33LVNm+5Rsn

Length: 6 -> 32 chars

[1] Encrypt [2] Decrypt [3] Exit
Select Option: 2
Enter text: NRu>0eCw$k/},oMzWUI@^33LVNm+5Rsn
Enter Numeric Seed (617-digit number): 12345

--- DECRYPTED RESULT ---
4+)wBR

[1] Encrypt [2] Decrypt [3] Exit
Select Option: |
```

Figure 8: Test 2

6.3 Test 3

Test Name	Extended Charset Encryption Test
Objective	Verify that the cipher correctly handles spaces, numbers, punctuation, and special characters from the 91-character charset
Action	1. Encrypt plaintext "Hello, World! 123 @\$" with seed "555666" 2. Decrypt the cipher text using the same seed 3. Compare all characters including spaces and symbols
Expected Result	Decrypted text preserves all spaces, punctuation, numbers, and special characters exactly: "Hello, World! 123 @\$"
Actual Result	Decrypted text: "Hello, World! 123 @\$" (all characters preserved)

Test Status	Successful
-------------	------------

Table 3: Test 3

Images below provide evidence:

```

C:\Users\Nitro v\AppData\Loc x + v
◆ GIOCO D'AZZARDO ◆
Multi-Round Serial Cipher (16 Rounds)

[1] Encrypt [2] Decrypt [3] Exit
Select Option: 1
Enter text: Hello, World! 123 @$
Enter Numeric Seed (617-digit number): 555666

--- ENCRYPTED RESULT ---
:=0k]NLU<Ie]Ny`n]s2h['K?9Gda+QXu]k>tsH[&@=.yiON#n#/Y9|:9gf:1s`5p"ehIS<PL`6/<'IRWC1EBu(H5p5{2;^rWYBZubY[
Length: 21 -> 103 chars

[1] Encrypt [2] Decrypt [3] Exit
Select Option: 2
Enter text: :=0k]NLU<Ie]Ny`n]s2h['K?9Gda+QXu]k>tsH[&@=.yiON#n#/Y9|:9gf:1s`5p"ehIS<PL`6/<'IRWC1EBu(H5p5{2;^rWYBZubY[
Enter Numeric Seed (617-digit number): 555666

--- DECRYPTED RESULT ---
Hello, World! 123 @$

[1] Encrypt [2] Decrypt [3] Exit
Select Option: |

```

Figure 9:Test 3

6.4 Test 4

Test Name	Noise Padding Randomness Test
Objective	Verify that encrypting the same plaintext multiple times with the same seed produces different cipher texts due to random noise padding
Action	1. Encrypt "TEST" with seed "777888" (First encryption) 2. Encrypt "TEST" with seed "777888" again (Second encryption) 3. Compare both cipher texts for differences in content and length

Expected Result	Two cipher texts are different in both content and possibly length due to random noise padding (2-5 characters per real character)
Actual Result	Cipher text 1: " zp0; (@%_911zZb.C5a=}C" Cipher text 2: " mK6#T(IO_I40v.baS5/tgp" Different content and length
Test Status	Successful

Table 4: Test 4

Images below provide evidence:

```

C:\Users\Nitro v\AppData\Loc  x  +  v
◆ GIOCO D'AZZARDO ◆
Multi-Round Serial Cipher (16 Rounds)

[1] Encrypt  [2] Decrypt  [3] Exit
Select Option: 1
Enter text: TEST
Enter Numeric Seed (617-digit number): 777888

--- ENCRYPTED RESULT ---
zp0;|(@%_911zZb.C5a=}C
Length: 4 -> 22 chars

[1] Encrypt  [2] Decrypt  [3] Exit
Select Option: 1
Enter text: TEST
Enter Numeric Seed (617-digit number): 777888

--- ENCRYPTED RESULT ---
mK6#T(IO_I40v.baS5/tgp
Length: 4 -> 22 chars

[1] Encrypt  [2] Decrypt  [3] Exit
Select Option: |

```

Figure 10: Test 4

6.5 Test 5

Test Name	Extended Seed Length Test (617-digit Support)
Objective	Verify that the cipher can handle very large numeric seed up to 100 digits without errors
Action	1. Encrypt "DATA" with a large numeric seed 2. Decrypt using the same large seed 3. Verify decryption accuracy
Expected Result	Two cipher texts are different in both content and possibly length due to random noise padding (2-5 characters per real character)
Actual Result	Cipher processes the large seed correctly; decrypted text matches original "DATA"
Test Status	Successful

Table 5: Test 5

Images below provide evidence:

Actual Result	Original length and Decrypted text matches exactly
Test Status	Successful

Table 6: Test 6

Images below provide evidence:

```

C:\Users\Witro v\AppData\Local\
+
-
GIOCO D'AZZARDO
Multi-Round Serial Cipher (16 Rounds)

[1] Encrypt [2] Decrypt [3] Exit
Select Option: 1
Enter text: Success is not a destination but a continuous journey of learning and adaptation. Every challenge you face is an opportunity to refine your skills and build resilience. Keep pushing forward with curiosity, as your unique perspective and consistent effort are the very things that will eventually turn your smallest goals into your greatest achievements
Enter Numeric Seed (617-digit number): 999000

--- ENCRYPTED RESULT ---
MgD[#nCj:J*-c;=*l#JdQf]bq9y<m>\}.oUy}h$[aeT.?Vj&9nhL]qk+'C3*5x[cn.(fu'3=.JKbeSLic<."6KT'oM,tL4w&OT3:0@hfaI.Xw+)#B5xro4VrdgD86:/`Rd@{R5[o4UD0utl&bJb[LWHx"Ik4PAp>b0=l{j$z/LqW6<TFnL2@Q}6C2IjR8%e$*1XCM@GJLyF`G4P;bD?G$FVOM(iG8hio2mb"9!DZQ\N@-`I"0{g2`&t`0a"s9B&$/=/g[=P=ni3|Krf 5j{0H-TN-0VUx@ix>!Vr,HMGI _:H(CX (o8<EA8h*"-+tb[,QbZQJ#<x*f4$ 7`k#]N_L-UwH[\rX7suGg[dV$dva]l8V)b3&I13:DIZ`Ms\5n_8vL%0}0omB1Z]r8V]YfT*,>,*t5H*PW-%}s:AALqhV'7,B);:a"|PCm[<[v>R&'p]Ps0A+h=[r]UNOG(3A!%&HqDnLH*760w;r'{{tstB:)}Ir>coH_CNIZ{? ka9$cfF',+3R5Hx?BGH_K-2IU|3<7*GoTA*`iJr"?:"s7c32{qQq!oIm%HwuT/x_?d;['f%;Ino-9INL*&.fjVr:.#ME8/>tpFV7.Xik*)Q]d|KcSR$75b|}$f;W0$!-R{?0.>>YQ0[$/kg9?*.>g|WzLoMGr*#0}okMP0`X=LS(/`YawWQ{[\npfo)bwp|bk`mGw0,7bX=-2UuePc1IQxL@&-_3|d)%$sbtm_kw*N|0UE{zokof1E`{z?|NM6i28>;Bny1XM$Y:Ns=-1;>YgU070-PqGmVL-f:w`q,|:WGI[-7Cv\DS=h0i ZV\+0&v+Ty(|m+2#GwJID94TP30hB.JJo8#c;k;&`SXL7R>*`DSESH`n;2<}V0-FJ9].%0IA<wtD469N!|\M/8X(C`jfc<eva3L?2y:w2DKX%oH2 &Rop2kp)>R(ie~)zf6Rqnm&-}5&#><f`RRbVnRuk#K*)V0YTRx9rRle|Lj/`z~`-s.AU= m5o<-DHZnd>fGSIar\ LW7`Z`SdjTF.BslgJ+`I>D0htio`SH6W0M07k$QT|A<C(X)Hj=<?m!ft|E(8N X:[CzIWJcVp,0FR7U)R8XsCV?@ucq(>jR= "PL(x|,R.6!K5E_{uG|EMdF>pLs=Q|/vJ_ZS)n2GZ2L%Em0E/Mj[z%2=]<$9`Tc+8`|I0zm610&86mMG%hN<+H`M*(XAX\((Z SGP,WU?+3t`9_5>84yS`Wn[Ia#EddVguy7(+&)KTSLOwghwi>+bAm|<Bx$Ec\?>D:ak8XFmA(>-CZ((XWm:.:V):VR/t,;`%GVY%}8Ad;4j58>z`|Rtu3qijsh6mEtIR`0ly)%##8#]]HljdYh|p|<zDIO(\.5B`xN`D:KIM/'6xAf-WIhxO|)\7%(q)p70vCzmGw_3V2Lf00w,2/w{)c\Qe.tx428<jCzyL8<:_ [7M@uYg0N['|mGsiV* 9SH,F2-YR}|e9J9)>B1BBszu)"bL_uw+/d:9vrrlQ!MY%2Idi+T,\K$K0$CfS&>|5+o,;`H$L`C3-eSP@x\QAnt0\@0tffk

Length: 352 -> 1575 chars

[1] Encrypt [2] Decrypt [3] Exit
Select Option: 2
Enter text: MgD[#nCj:J*-c;=*l#JdQf]bq9y<m>\}.oUy}h$[aeT.?Vj&9nhL]qk+'C3*5x[cn.(fu'3=.JKbeSLic<."6KT'oM,tL4w&OT3:0@hfaI.Xw+)#B5xro4VrdgD86:/`Rd@{R5[o4UD0utl&bJb[LWHx"Ik4PAp>b0=l{j$z/LqW6<TFnL2@Q}6C2IjR8%e$*1XCM@GJLyF`G4P;bD?G$FVOM(iG8hio2mb"9!DZQ\N@-`I"0{g2`&t`0a"s9B&$/=/g[=P=ni3|Krf 5j{0H-TN-0VUx@ix>!Vr,HMGI _:H(CX (o8<EA8h*"-+tb[,QbZQJ#<x*f4$ 7`k#]N_L-UwH[\rX7suGg[dV$dva]l8V)b3&I13:DIZ`Ms\5n_8vL%0}0omB1Z]r8V]YfT*,>,*t5H*PW-%}s:AALqhV'7,B);:a"|PCm[<[v>R&'p]Ps0A+h=[r]UNOG(3A!%&HqDnLH*760w;r'{{tstB:)}Ir>coH_CNIZ{? ka9$cfF',+3R5Hx?BGH_K-2IU|3<7*GoTA*`iJr"?:"s7c32{qQq!oIm%HwuT/x_?d;['f%;Ino-9INL*&.fjVr:.#ME8/>tpFV7.Xik*)Q]d|KcSR$75b|}$f;W0$!-R{?0.>>YQ0[$/kg9?*.>g|WzLoMGr*#0}okMP0`X=LS(/`YawWQ{[\npfo)bwp|bk`mGw0,7bX=-2UuePc1IQxL@&-_3|d)%$sbtm_kw*N|0UE{zokof1E`{z?|NM6i28>;Bny1XM$Y:Ns=-1;>YgU070-PqGmVL-f:w`q,|:WGI[-7Cv\DS=h0i ZV\+0&v+Ty(|m+2#GwJID94TP30hB.JJo8#c;k;&`SXL7R>*`DSESH`n;2<}V0-FJ9].%0IA<wtD469N!|\M/8X(C`jfc<eva3L?2y:w2DKX%oH2 &Rop2kp)>R(ie~)zf6Rqnm&-}5&#><f`RRbVnRuk#K*)V0YTRx9rRle|Lj/`z~`-s.AU= m5o<-DHZnd>fGSIar\ LW7`Z`SdjTF.BslgJ+`I>D0htio`SH6W0M07k$QT|A<C(X)Hj=<?m!ft|E(8N X:[CzIWJcVp,0FR7U)R8XsCV?@ucq(>jR= "PL(x|,R.6!K5E_{uG|EMdF>pLs=Q|/vJ_ZS)n2GZ2L%Em0E/Mj[z%2=]<$9`Tc+8`|I0zm610&86mMG%hN<+H`M*(XAX\((Z SGP,WU?+3t`9_5>84yS`Wn[Ia#EddVguy7(+&)KTSLOwghwi>+bAm|<Bx$Ec\?>D:ak8XFmA(>-CZ((XWm:.:V):VR/t,;`%GVY%}8Ad;4j58>z`|Rtu3qijsh6mEtIR`0ly)%##8#]]HljdYh|p|<zDIO(\.5B`xN`D:KIM/'6xAf-WIhxO|)\7%(q)p70vCzmGw_3V2Lf00w,2/w{)c\Qe.tx428<jCzyL8<:_ [7M@uYg0N['|mGsiV* 9SH,F2-YR}|e9J9)>B1BBszu)"bL_uw+/d:9vrrlQ!MY%2Idi+T,\K$K0$CfS&>|5+o,;`H$L`C3-eSP@x\QAnt0\@0tffk

```

Figure 12:Test 6



Figure 13: Test 6

6.7 Test 7

Test Name	Upper and Lowercase Letter Test
Objective	Verify that the cipher preserves the exact case of letters (uppercase vs lowercase) throughout encryption and decryption
Action	1. Encrypt mixed-case text "AbCdEfGhIj" with seed "111222" 2. Decrypt the cipher text 3. Verify that case is preserved exactly
Expected Result	Decrypted text maintains exact case pattern: "AbCdEfGhIj"

Actual Result	Decrypted text: "AbCdEfGhIj" (case preserved perfectly)
Test Status	Successful

Table 7: Test 7

Images below provide evidence:

```

C:\Users\Nitro v\AppData\Loc  X  +  v
  ◆ GIOCO D'AZZARDO ◆
Multi-Round Serial Cipher (16 Rounds)

[1] Encrypt [2] Decrypt [3] Exit
Select Option: 1
Enter text: AbCdEfGhIj
Enter Numeric Seed (617-digit number): 111222

--- ENCRYPTED RESULT ---
pcB.4;=/80$mjGsSNL~Q9'p'dC8Vu'VT\is$Iv gX8SZh\p9p-
Length: 10 -> 50 chars

[1] Encrypt [2] Decrypt [3] Exit
Select Option: 2
Enter text: pcB.4;=/80$mjGsSNL~Q9'p'dC8Vu'VT\is$Iv gX8SZh\p9p-
Enter Numeric Seed (617-digit number): 111222

--- DECRYPTED RESULT ---
AbCdEfGhIj

[1] Encrypt [2] Decrypt [3] Exit
Select Option: |

```

Figure 14:Test 7

7 Critical Evaluation of New Cipher

7.1 Strengths

7.1.1 Elimination of Frequency Analysis Attacks

Analysis: The algorithm successfully defeats traditional frequency analysis, which is the primary weakness of classical substitution ciphers.

How it works:

- Noise padding adds 2-5 random characters before each real character
- The same plaintext character encrypts differently based on its position
- Random noise makes statistical analysis impossible

Impact: Unlike the Caesar cipher where 'E' always maps to the same letter, this cipher produces different outputs for the same input character, making frequency analysis completely ineffective. For example, the word "HELLO" might encrypt the two 'L' characters completely differently.

7.1.2 Massive Key Space

Analysis: The key space is astronomically large compared to Caesar cipher's trivial 25 possible keys.

Mathematical comparison:

- Caesar cipher: 25 possible keys
- This cipher: Supports up to 617-digit numeric seeds
- Key space: $\sim 10^{617}$ possibilities (virtually unlimited)

Impact: Brute-force attacks become computationally infeasible. Even testing 1 billion keys per second, it would take longer than the age of the universe to exhaust even a fraction of the key space. This makes the cipher resistant to brute-force attacks, which would break Caesar cipher in milliseconds.

7.1.3 Avalanche Effect

Analysis: The 16-round serial transformation creates a strong avalanche effect where small input changes cascade into dramatically different outputs.

Mechanism:

- Each round reshuffles the entire 91-character charset
- Applies random directional shifts (left/right)
- Uses random shift magnitudes (1-6 positions)
- Each round's output becomes the next round's input

Impact: Changing a single bit in the plaintext or seed results in completely different cipher text. This is a critical property of strong cryptographic algorithms. For instance, encrypting "HELLO" vs "HALLO" with the same seed produces entirely unrelated cipher texts, preventing pattern-based attacks.

7.1.4 Position-Dependent Encryption

Analysis: Each character's encryption depends on its position in the message, preventing identical plaintexts from producing identical cipher texts.

Implementation:

Character PRNG = Random(base_seed + position)

Impact: The word "THE" appearing at position 0 encrypts completely differently than "THE" at position 50, even with the same seed. This prevents attackers from identifying repeated words or phrases, which is a common cryptanalysis technique. This addresses a major weakness in classical ciphers where patterns remain visible.

7.1.5 Non-Deterministic Cipher text

Analysis: The noise padding mechanism ensures that encrypting the same message multiple times produces different cipher texts, even with identical seeds.

Randomness sources:

- 2-5 random noise characters before each real character
- 2-5 random noise characters at the end
- Noise uses Python's `random.choice()` with system entropy

Impact: This prevents known-plaintext attacks where attackers compare multiple encryptions of the same message. An attacker intercepting multiple encryptions of "ATTACK AT DAWN" cannot confirm they're the same message. This property, called "probabilistic encryption," is found in modern cryptographic systems like RSA-OAEP.

7.2 Weakness

7.2.1 Weak Pseudorandom Number Generator (PRNG)

Critical Issue: The algorithm uses Python's `random.Random()`, which is a Mersenne Twister PRNG designed for statistical simulations, NOT cryptographic security.

Vulnerabilities:

- Mersenne Twister is predictable if an attacker observes enough outputs
- State can be recovered from 624 consecutive outputs
- Not designed to be cryptographically secure

Impact: A sophisticated attacker who intercepts multiple cipher texts could potentially predict the PRNG sequence and reverse-engineer the encryption.

7.2.2 Cipher text Expansion

Critical Issue: The noise padding causes significant cipher text expansion, making the encrypted message 3-6 times larger than the original.

Example:

- Plaintext: "HELLO" (5 characters)
- Cipher text: ~20-30 characters (4-6x expansion)
- 1 MB file → 3-6 MB encrypted file

Problems:

- Increased bandwidth consumption for network transmission
- Higher storage requirements
- Slower processing for large files
- Makes it impractical for resource-constrained environments (IoT devices, embedded systems)

Impact: For a 100 MB document, encryption could produce a 300-600 MB file, making it unsuitable for applications with strict size constraints like SMS encryption or low-bandwidth communications.

7.2.3 Lack of Authentication and Integrity Verification

Critical Issue: The cipher provides confidentiality only, with no mechanism to verify message integrity or authenticity.

Missing features:

- No Message Authentication Code (MAC)

- No digital signature
- No way to detect if cipher text was tampered with
- No protection against replay attacks

Impact: An attacker could:

- Modify the cipher text without detection
- Replay old messages
- Perform bit-flipping attacks
- Insert malicious content

Real-world scenario: If Alice sends "TRANSFER \$100" encrypted, an attacker could modify the cipher text bits. Upon decryption, Bob might receive garbled text or potentially manipulated amounts, with no way to verify the message wasn't altered.

7.3 Application Area

1. Educational and Academic Environments

Suitable for: Teaching cryptography concepts, student projects, cryptanalysis training, and demonstrating multi-round encryption principles.

Justification: The algorithm effectively demonstrates advanced concepts like avalanche effect and noise padding while remaining understandable for learning purposes.

2. Personal Messaging and Document Protection (Low-Stakes)

Suitable for: Personal diary encryption, private notes, casual messaging between friends, and password-protecting non-sensitive documents.

Justification: Provides reasonable protection against casual attackers for non-critical personal data. NOT suitable for banking, medical records, or sensitive information.

3. Gaming and Entertainment Software

Suitable for: Protecting game save files, encrypting in-game communications, anti-cheat mechanisms, and protecting game assets from casual modification.

Justification: Gaming applications don't require military-grade encryption. This cipher provides sufficient protection against casual players while maintaining acceptable performance.

4. Internal Corporate Communications (Non-Confidential)

Suitable for: Internal memos, non-confidential project documentation, casual inter-office messaging, and draft documents.

Justification: Provides a balance between security and simplicity for internal communications. Diffie-Hellman compatibility allows easy key exchange. Should NOT be used for financial data, trade secrets, or regulatory-compliant information.

5. CTF Competitions and Security Training

Suitable for: Cybersecurity competitions, penetration testing training, cryptanalysis challenges, and security awareness demonstrations.

Justification: Perfect for training scenarios where participants must analyse and potentially break custom encryption schemes. Complex enough to be challenging but has identifiable weaknesses for educational exploitation.

Appropriate Use: Educational purposes, low-stakes personal applications, and as an additional obfuscation layer in multi-layer security systems.

Inappropriate Use: Financial systems, healthcare data, legal documents, military communications, or any application subject to regulatory compliance.

8 Conclusion

This coursework successfully developed the Gioco d'azzardo Cipher, addressing fundamental Caesar cipher vulnerabilities through mathematical enhancements (modular arithmetic, position-based seed derivation, 617-digit key support) and logical modifications (16-round transformation, Fisher-Yates shuffling, noise padding). Comprehensive testing validated functional correctness, security through wrong-seed rejection, extended character set handling, and non-deterministic cipher text generation. Critical evaluation identified significant strengths including frequency analysis elimination, massive key space (10^{617} possibilities), and avalanche effects, while acknowledging limitations such as non-cryptographic PRNG reliance, cipher text expansion, and authentication absence. The algorithm demonstrates practical applicability in educational environments, gaming software, and low-stakes applications, while remaining unsuitable for high-security scenarios requiring regulatory compliance. This project successfully illustrates how classical cryptographic concepts can be systematically enhanced to address modern security requirements while maintaining educational clarity and computational efficiency.

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